Graph Review. Let $G = (V, E)$ be a connected graph with edge weights. Suppose G is undirected and represented by adjacency lists. Recall the time complexities of:

1. DFS/BFS:
2. Prim's algorithm:
3. Kruskal's algorithm:

Lazy Updates to the MST. Let $G = (V, E)$ be a connected graph with distinct edge weights, and let T be its minimum spanning tree (MST). Graphs are undirected and represented by adjacency lists. In each part, edge weights change, but the modified weights are still distinct.

1. The weight of one edge $e \notin T$ decreases. Find the new MST in $O(|V|)$ time.
2. The weight of one edge $e \in T$ increases by an arbitrary positive amount. Find the new MST in $O(|V| + |E|)$ time.
3. Fix a vertex v and let d be its degree in T . The weight of every edge incident to v increases by exactly 1. Find the new MST in $O(|V| + |E| + d^2)$ time.

Dishan's MST with Small Weight. Suppose $G = (V, E)$ is connected and every edge weight belongs to $\{6, 7, 67\}$. Design a $O(|V| + |E|)$ time algorithm to find an MST of G.

Aren't you Glad the Constant isn't s*xs*v*n. Suppose $G = (V, E)$ is connected and every edge weight is a positive integer bounded by 381. Give an efficient algorithm to find the MST in G.